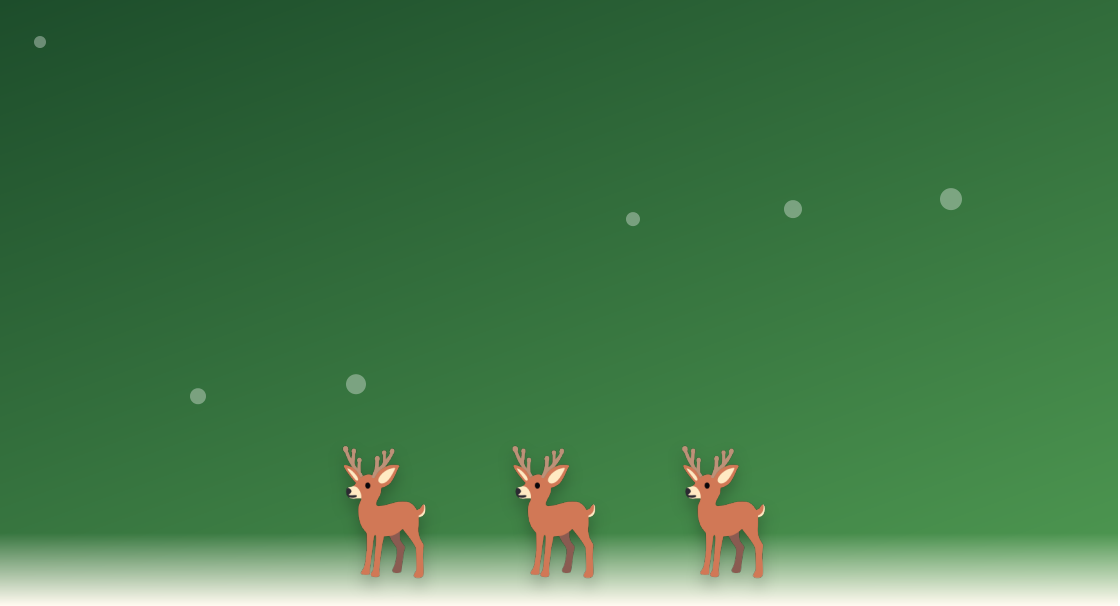


The Population Party

II PUC · Ch 11 · Organisms and Populations

Starring Bindu 👧 & Prof. BAIO 🦉

A BAIO storybook by Narendra Sir · Biology Lecturer



Who's at the Party?

Prof. BAIO pointed at a herd of deer. "A **population** is a group of the same species living in one area. It has features an individual doesn't: a **birth rate**, a **death rate**, a **sex ratio** and an **age pyramid**."

> 💡 **REMEMBER**
> Age pyramids: triangular — expanding · bell-shaped — stable · urn-shaped — declining.



Counting Heads

Population **density** can be counted directly, measured as percent cover, or estimated from signs like tiger pug marks. It changes by four processes: **natality** and **immigration** add individuals; **mortality** and **emigration** remove them.

💡 **REMEMBER**

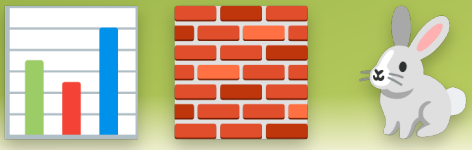
$N_{t+1} = N_t + [(B + I) - (D + E)]$.



Growing Fast

With unlimited food and space, a population grows faster and faster — **exponential growth**, a J-shaped curve: dN/dt = rN. Here r is the **intrinsic rate of natural increase**.

> 💡 **REMEMBER**
> r for the Norway rat is 0.015; for the flour beetle 0.12; for India in 1981, 0.0205.



Hitting the Ceiling

In nature, resources run short. Growth slows and levels off at the **carrying capacity**, K. This S-shaped curve is **logistic growth** (Verhulst–Pearl): $dN/dt = rN(K - N)/K$.

💡 **REMEMBER**

Logistic growth is more realistic than exponential growth.



Hunters and Hunted

Predation keeps prey in check. In Australia, prickly pear cactus spread wildly until a cactus-eating moth was brought in. Prey defend themselves: the **Monarch butterfly** tastes bad to birds, and **Calotropis** makes poisonous cardiac glycosides.

> 💡 **REMEMBER**
> **Nicotine, caffeine, quinine and opium are plant chemicals that defend against grazers.**



Rivals

When species need the same resource, there is **competition**. On the Galápagos Islands, goats arrived and the Abingdon tortoise disappeared within ten years. **Gause** said two closely related species competing for the same resource cannot live together for ever — **competitive exclusion**.

💡 REMEMBER

Resource partitioning (like MacArthur's warblers) lets competitors coexist.



Freeloaders

In **parasitism**, one gains and the other loses. Cuscuta, a leafless plant, sucks food from its host. The cuckoo lays its eggs in a crow's nest — **brood parasitism**. Lice on humans are ectoparasites.

💡 **REMEMBER**

Many parasites are host-specific and have complex life cycles.



Best Friends

In **commensalism**, one gains and the other is unaffected — an orchid on a mango branch, barnacles on a whale. In **mutualism**, both gain — lichens, mycorrhiza, the fig and its wasp. The orchid **Ophrys** even tricks male bees into pollinating it — **sexual deceit**.

💡 **REMEMBER**

Mutualism (+, +) · Commensalism (+, 0) · Amensalism (−, 0) · Competition (−, −) · Predation and parasitism (+, −).



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

population p.1 · birth rate p.1 · death rate p.1 · sex ratio p.1

age pyramid p.1 · density p.2 · natality p.2 · immigration p.2

mortality p.2 · emigration p.2 · exponential growth p.3

intrinsic rate of natural increase p.3 · carrying capacity p.4

logistic growth p.4 · Predation p.5 · Monarch butterfly p.5

Calotropis p.5 · competition p.6 · Gause p.6

competitive exclusion p.6 · parasitism p.7 · brood parasitism p.7

commensalism p.8 · mutualism p.8 · Ophrys p.8 · sexual deceit p.8



QUICK QUIZ

Do you remember?

1. Logistic growth gives a curve that is:

- J-shaped
- S-shaped
- Bell-shaped
- Straight

2. An urn-shaped age pyramid shows a population that is:

- Growing
- Stable
- Declining
- Exploding

3. Cuckoo laying eggs in a crow's nest is:

- Commensalism
- Brood parasitism
- Mutualism
- Predation

4. The competitive exclusion principle was given by:

- Darwin
- Gause
- Verhulst
- MacArthur

5. An orchid growing on a mango branch shows:

- Parasitism
- Mutualism
- Commensalism
- Amensalism

Answers: 1-S-shaped · 2-Declining · 3-Brood parasitism · 4-Gause · 5-Commensalism